

AI RESUME ANALYZER & JOB MATCHER



MINI PROJECT

21CS303-Artificial Intelligence and Machine Learning Laboratory

(for the Academic Year 2026-27)

Done by

ASWIN KUMAR T A – 913124104302

Department of Computer Science and Engineering

VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY

VIRAGANOOR, MADURAI-625009

(AUTONOMOUS)



**VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY,
MADURAI-625009.**

(Autonomous)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



2025-26

BONAFIDE CERTIFICATE

This is to certify that the mini project work titled

AI RESUME ANALYZER & JOB MATCHER

Was done by

ASWIN KUMAR T A – 913124104302

of fourth semester –Computer Science and Engineering Department in partial fulfillment of the requirement of 21CS303-Artificial Intelligence and Machine Learning Laboratory during the year 2025-26.

Course in-charge

EX.NO: 10

AI RESUME ANALYZER & JOB MATCHER

DATE: 26/09/26

AIM:

To design and develop an AI-powered Resume Analyzer and Job Matcher that uses Natural Language Processing and Machine Learning techniques to analyze resumes, extract candidate skills, calculate resume quality, and match students with suitable job opportunities while providing separate interfaces for students and companies.

CONCEPTS INVOLVED:

- **Artificial Intelligence and Machine Learning (AI/ML):**

Provides automated intelligence for career readiness, resume scoring, and profile-job vector matching.

- **Natural Language Processing (NLP):**

Used to clean, tokenize, and extract meaningful semantic textual patterns from unstructured resume files and job descriptions.

- **Resume Text Extraction:**

Multi-format document parsing engine supporting PDF, DOCX, and TXT using PyPDF and python-docx.

- **Skill Extraction Taxonomy:**

Hierarchical domain taxonomy matching candidate skills across Web, Mobile, Backend, Database, Cloud, and AI categories.

- **TF-IDF Vectorization:**

Transforms document corpus into statistical Term Frequency-Inverse Document Frequency numerical feature vectors.

- **Cosine Similarity Metric:**

Calculates angular cosine distance between candidate resume vectors and employer job vectors to produce an accurate match percentage.

- **Resume Quality & ATS Health Scoring:**

Evaluates section completeness, formatting hygiene, keyword density, and technical depth with explainable suggestions.

- **Role-Based Access Control (RBAC):**

Ensures isolated security, authentication, and tailored dashboards for Students, Employers, and Administrators.

- **RESTful API Architecture (Flask Backend):**

Handles server-side business logic, JWT authentication, NLP execution pipelines, and database interactions.

- **Relational Database Management (MySQL & XAMPP):**

Maintains ACID-compliant structured storage for users, profiles, resumes, skills, job postings, and applications.

FILES USED:

Frontend (Vue 3 + Vite):

- frontend/src/views/public/LandingView.vue
- frontend/src/views/public/LoginView.vue
- frontend/src/views/public/RegisterView.vue
- frontend/src/views/student/StudentDashboardView.vue
- frontend/src/views/student/ResumeUploadView.vue
- frontend/src/views/student/SkillsView.vue

Backend (Python + Flask):

- backend/app.py
- backend/config.py
- backend/database/db.py
- backend/routes/auth_routes.py
- backend/routes/resume_routes.py
- backend/routes/job_routes.py
- backend/routes/application_routes.py

AI / ML & NLP Engine:

- backend/ai/resume_parser.py
- backend/ai/skill_extractor.py
- backend/ai/matcher.py
- backend/ai/skills_taxonomy.py

Database & Schema:

- database/schema.sql
- backend/init_db.py

PROGRAM:

db.py (Database Connection & MySQL Pool):

```
import pymysql
from dbutils.pooled_db import PooledDB
from config import Config

pool = PooledDB(
    creator=pymysql,
    maxconnections=10,
    mincached=2,
    host=Config.DB_HOST,
    port=Config.DB_PORT,
    user=Config.DB_USER,
    password=Config.DB_PASSWORD,
    database=Config.DB_NAME,
    cursorclass=pymysql.cursors.DictCursor,
    autocommit=True
)

def get_db():
    return pool.connection()
```

auth_routes.py (Authentication & JWT Token Generation):

```
from flask import Blueprint, request, jsonify
from werkzeug.security import generate_password_hash, check_password_hash
import jwt, datetime
from config import Config
from database.db import get_db

auth_bp = Blueprint('auth', __name__)

@auth_bp.route('/register', methods=['POST'])
```

```

def register():
    data = request.get_json() or {}
    name = data.get('name', '').strip()
    email = data.get('email', '').strip().lower()
    password = data.get('password', '')
    role = data.get('role', 'student')

    if not name or not email or not password:
        return jsonify({"success": False, "message": "Missing required fields"}), 400

    pwd_hash = generate_password_hash(password)
    conn = get_db()
    with conn.cursor() as cursor:
        cursor.execute(
            "INSERT INTO users (name, email, password_hash, role) VALUES (%s, %s, %s, %s)",
            (name, email, pwd_hash, role)
        )
        user_id = cursor.lastrowid
        if role == 'student':
            cursor.execute("INSERT INTO student_profiles (user_id, college, degree) VALUES (%s, %s, %s)",
                           (user_id, data.get('college'), data.get('degree')))
        elif role == 'company':
            cursor.execute("INSERT INTO company_profiles (user_id, company_name) VALUES (%s, %s)",
                           (user_id, name))
    return jsonify({"success": True, "message": "Registered successfully"}), 201

```

resume_parser.py (Multi-Format Resume Text Extraction):

```

import os
from pypdf import PdfReader
import docx

def extract_text_from_file(file_path, file_type):
    file_type = file_type.lower()
    if file_type == 'pdf':
        text = ""
        reader = PdfReader(file_path)
        for page in reader.pages:
            t = page.extract_text()
            if t: text += t + "\n"
        return text.strip()
    elif file_type in ['docx', 'doc']:
        doc = docx.Document(file_path)
        text = "\n".join([p.text for p in doc.paragraphs if p.text.strip()])
        return text.strip()
    elif file_type == 'txt':
        with open(file_path, 'r', encoding='utf-8', errors='ignore') as f:
            return f.read().strip()
    return ""

```

skill_extractor.py (Taxonomy-Based NLP Skill Extraction):

```

import re
from ai.skills_taxonomy import SKILLS_TAXONOMY

def extract_skills_from_text(raw_text):
    clean_text = raw_text.lower()
    extracted_skills = []

    for category, skill_dict in SKILLS_TAXONOMY.items():
        for canonical_skill, aliases in skill_dict.items():
            pattern = r'\b' + re.escape(canonical_skill.lower()) + r'\b'
            if re.search(pattern, clean_text):
                extracted_skills.append({
                    "skill_name": canonical_skill,
                    "category": category,
                    "confidence": 0.95
                })
        continue
    for alias in aliases:

```

```

        if re.search(r'\b' + re.escape(alias.lower()) + r'\b', clean_text):
            extracted_skills.append({
                "skill_name": canonical_skill,
                "category": category,
                "confidence": 0.90
            })
        break
    return extracted_skills

```

matcher.py (TF-IDF & Cosine Similarity Job Matching Pipeline):

```

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

def calculate_job_compatibility(candidate_skills, resume_text, job_skills, job_description):
    cand_set = set([s.lower().strip() for s in candidate_skills])
    job_set = set([s.lower().strip() for s in job_skills])

    matched = cand_set.intersection(job_set)
    missing = job_set - cand_set

    # 1. Skill Match Ratio (60% weight)
    skill_score = (len(matched) / len(job_set)) * 100.0 if job_set else 70.0

    # 2. TF-IDF Cosine Semantic Similarity (40% weight)
    vectorizer = TfidfVectorizer(stop_words='english')
    corpus = [resume_text or " ".join(candidate_skills), job_description or " ".join(job_skills)]
    tfidf_matrix = vectorizer.fit_transform(corpus)
    cosine_sim = float(cosine_similarity(tfidf_matrix[0:1], tfidf_matrix[1:2])[0][0]) * 100.0

    overall_match = round(0.60 * skill_score + 0.40 * cosine_sim, 1)

    return {
        "match_score": min(100.0, max(0.0, overall_match)),
        "matched_skills": list(matched),
        "missing_skills": list(missing),
        "skill_score": round(skill_score, 1),
        "semantic_similarity": round(cosine_sim, 1)
    }

```

job_routes.py (Company Job Posting & AI Compatibility Evaluation):

```

from flask import Blueprint, request, jsonify
from database.db import get_db
from ai.matcher import calculate_job_compatibility

jobs_bp = Blueprint('jobs', __name__)

@jobs_bp.route('/create', methods=['POST'])
def create_job():
    data = request.get_json() or {}
    company_id = data.get('company_id')
    title = data.get('title')
    skills = data.get('skills')
    desc = data.get('description')

    conn = get_db()
    with conn.cursor() as cursor:
        cursor.execute(
            """INSERT INTO jobs
               (company_id, title, job_type, location, experience_required, skills, description,
                salary_min, salary_max)
               VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s)"""
            , (company_id, title, data.get('job_type', 'Full Time'), data.get('location'),
              data.get('experience_required'), skills, desc, data.get('salary_min'),
              data.get('salary_max'))
        )
        job_id = cursor.lastrowid
    return jsonify({"success": True, "job_id": job_id, "message": "Job published"}), 201

```

OUTPUT:

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Home

Find Jobs

How It Works

About

Sign In

Get Started Free

Create Your Account

Join the intelligent AI career and matching network

I am a Student / Job Seeker

I am an Employer / Company

Full Name *

ASWIN KUMAR

Email Address *

cseaswin@gmail.com

Password (Min 6 chars) *

.....

Confirm Password *

.....

College / University

ngineering and Technology

Degree & Field

e.g. B.S. in Computer Scien

Fig 1: Student Account Registration Interface

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Home

Find Jobs

How It Works

About

Sign In

Get Started Free

Welcome Back

Sign in to your AI Career & Recruitment account

Quick 1-Click Demo Logins:

Student

Company

Admin

Email Address

cseaswin@gmail.com

Password

.....

Sign In

Don't have an account?

Create an account

Fig 2: Student Authentication & Sign In

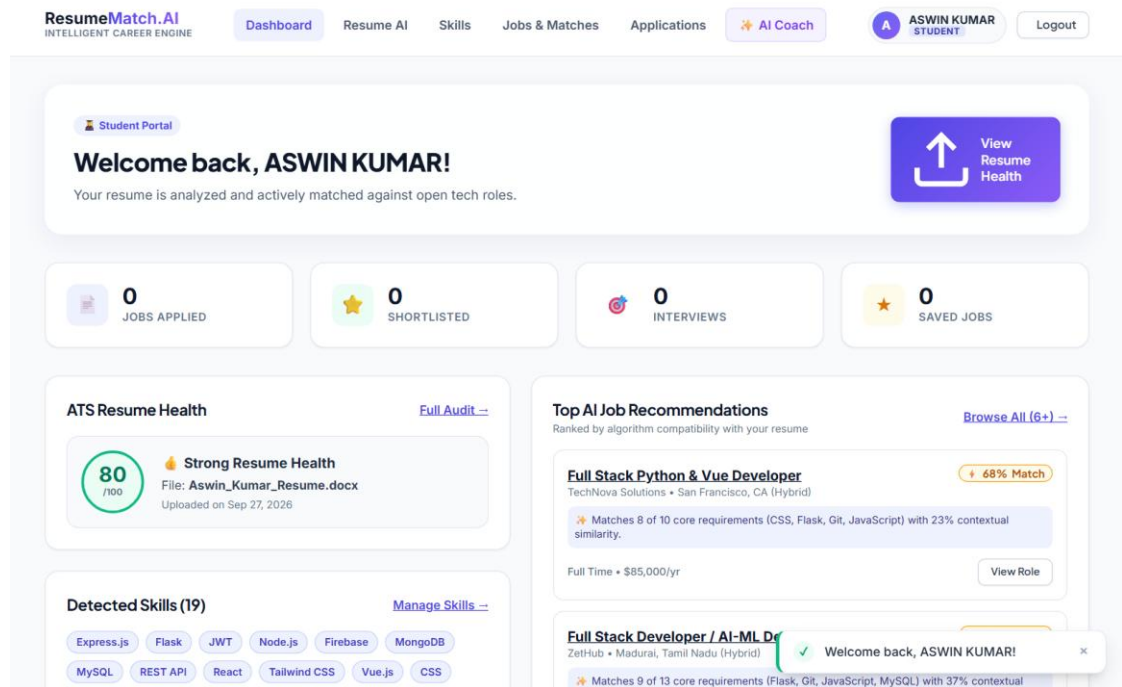


Fig 3: Student Interactive Dashboard

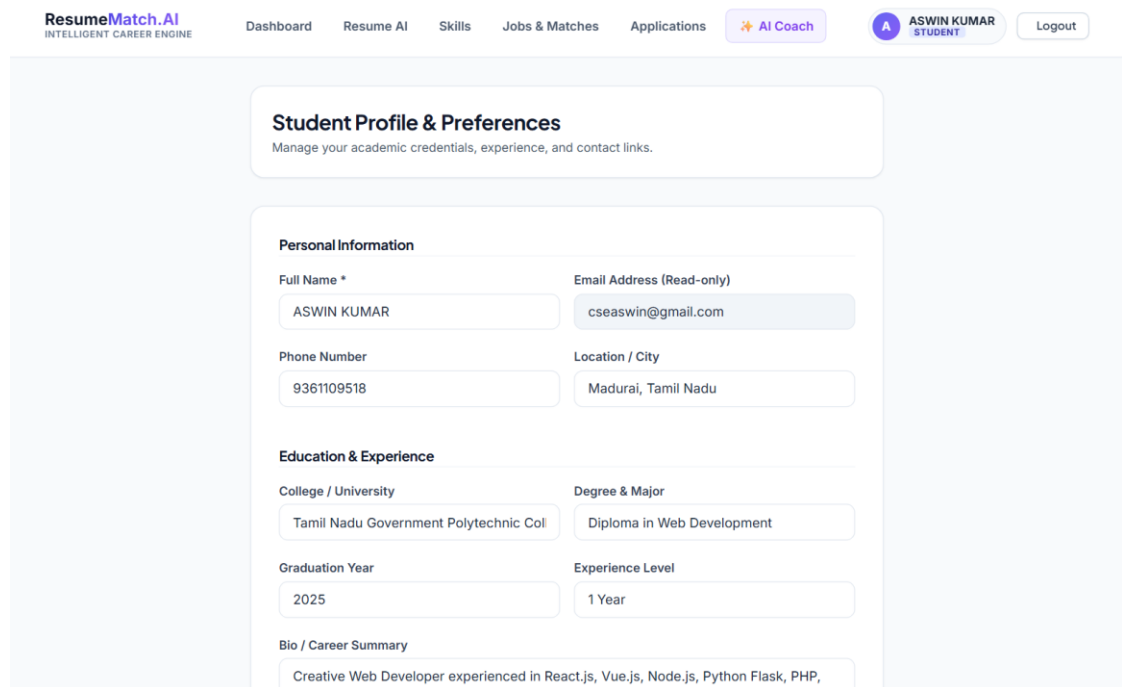


Fig 4: Student Technical Profile & Details

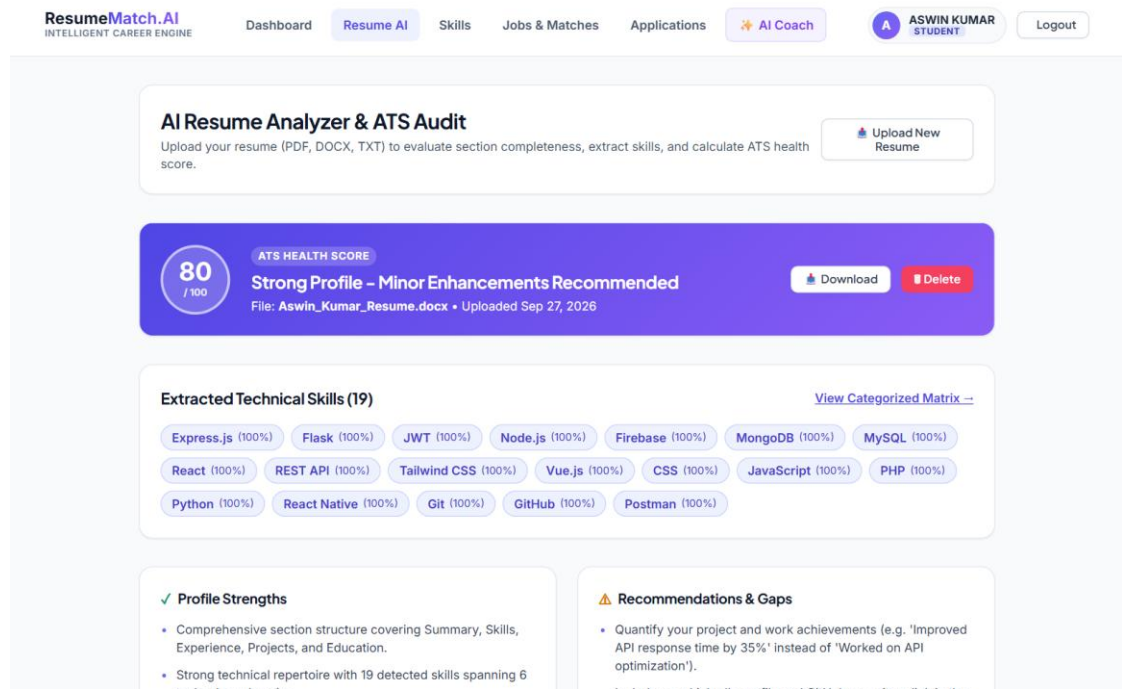


Fig 5: AI Resume Analyzer & ATS Audit Dropzone

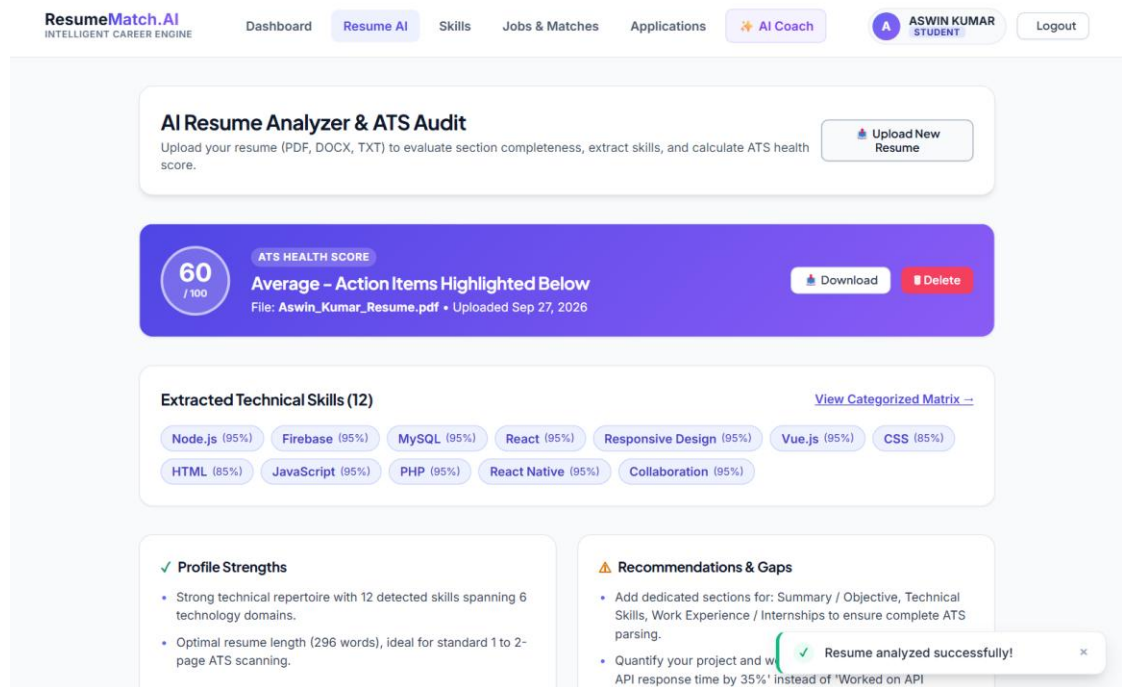


Fig 6: Resume Upload & Real-time NLP Processing

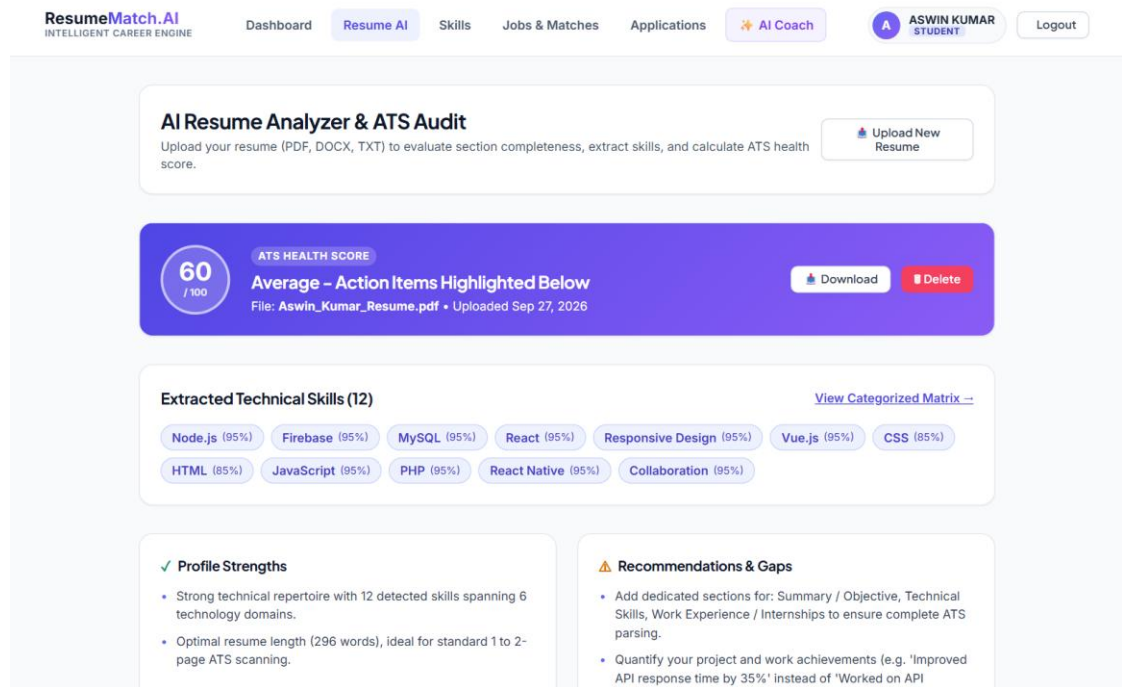


Fig 7: AI Resume Health Score & Comprehensive Audit

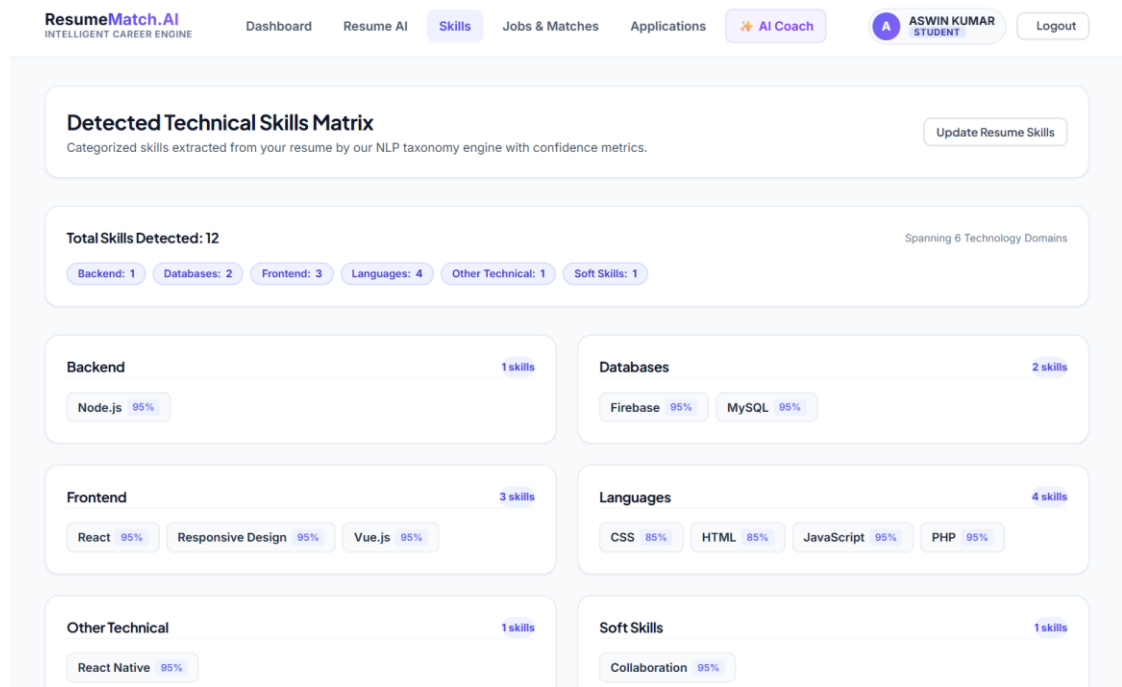


Fig 8: Extracted Candidate Technical Skills Matrix

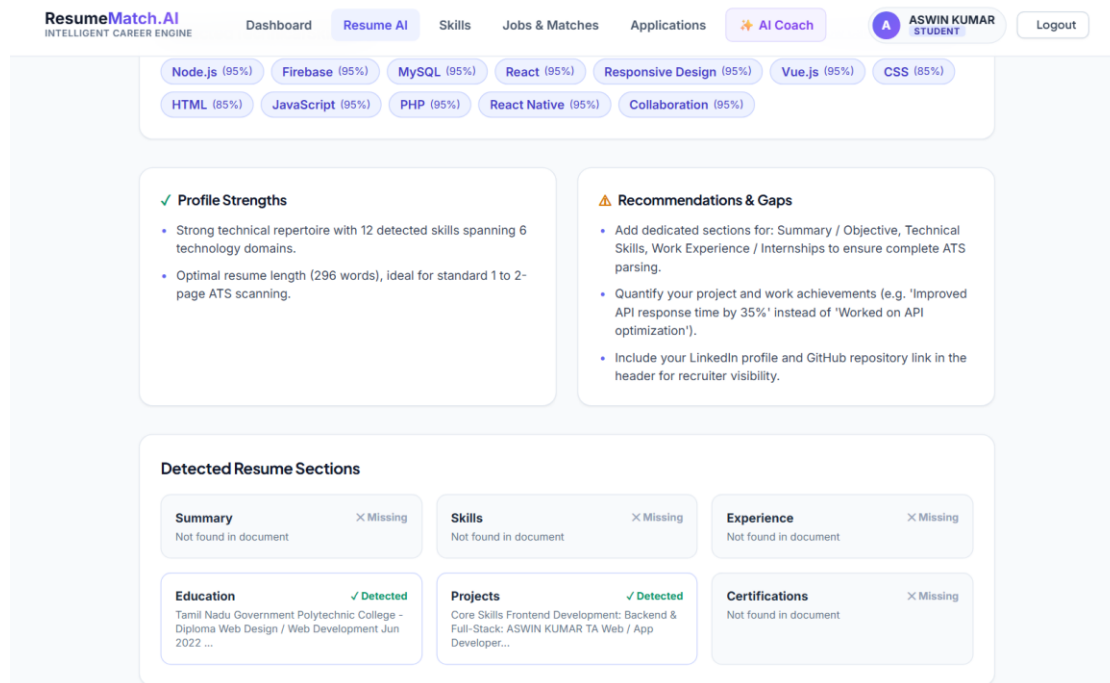


Fig 9: Resume ATS Quality Breakdown & Action Items

The screenshot shows the "Create Your Account" page on the ResumeMatch.AI website. The page has a header with the ResumeMatch.AI logo and navigation links: Home, Find Jobs, How It Works, and About. On the right side of the header are "Sign In" and "Get Started Free" buttons.

The main content area is titled "Create Your Account" with the subtitle "Join the intelligent AI career and matching network". Below this, there are two radio button options: "I am a Student / Job Seeker" (selected) and "I am an Employer / Company".

Below the radio buttons, there are four input fields arranged in two rows:

- Company Name * (ZetHub IT Solutions)
- Official Work Email * (hr@zethub.in)
- Password (Min 6 chars) * (masked with dots)
- Confirm Password * (masked with dots)

Fig 10: ZetHub Employer Account Registration

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Home

Find Jobs

How It Works

About

Sign In

Get Started Free

Welcome Back

Sign in to your AI Career & Recruitment account

Quick 1-Click Demo Logins:

Student

Company

Admin

Email Address

hr@zethub.in

Password

.....

Sign In

Don't have an account?

Create an account

Fig 11: ZetHub Company Sign In

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Dashboard

Manage Jobs

+ Post Job

Applicants

Profile

Z ZetHub COMPANY

Logout

Employer Portal

Employer Dashboard

Manage open positions, review AI-ranked candidates, and streamline your hiring pipeline.

+ Post New Job

0

ACTIVE JOBS

0

TOTAL APPLICANTS

0

SHORTLISTED

0%

AVG MATCH SCORE

Recent Candidate Applications

View All Applicants -->

No applicants received yet. Post jobs to start receiving AI-matched applications!

✓ Welcome back, ZetHub!

✕

Fig 12: ZetHub Recruitment Dashboard

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Dashboard

Manage Jobs

+ Post Job

Applicants

Profile

Z

ZetHub COMPANY

Logout

Required Skills (Comma-separated) *

React.js, Vue.js, Node.js, Python, Flask, I

Preferred Skills (Optional)

NLP, Scikit-learn, Electron.js, Firebase

Used as primary targets in the AI matching formula.

Min Annual Salary (\$)

400000

Max Annual Salary (\$)

750000

Application Deadline

dd-mm-yyyy

Full Job Description *

ZetHub IT Solutions is seeking a talented Full Stack & AI-ML Developer Intern. You will build cutting-edge web applications using Vue.js, React, Node.js, Python Flask, and MySQL, with integration to machine learning recommendation models.

Requirements & Responsibilities

Qualifications, academic background, preferred certifications, and day-to-day deliverables...

Listing Status

Active (Visible on J

Fig 13: Job Creation Form with Target Skills & Attributes

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Dashboard

Manage Jobs

+ Post Job

Applicants

Profile

Z

ZetHub COMPANY

Logout

Manage Job Postings

View active vacancies, applicant volumes, and manage listing statuses.

+ Post New Job

Job Title	Type & Location	Applicants	Status	Created Date	Actions
Full Stack Developer / AI-ML Developer Intern React.js, Vue.js, Node.js, Python, Flask, MySQL, JavaScript, HTML, CSS, REST API, Git	Full Time Madurai, Tamil Nadu (Hybrid / On-site)	0 applicants	active	Sep 27, 2026	EditDelete

Fig 14: ZetHub Active Job Postings Management

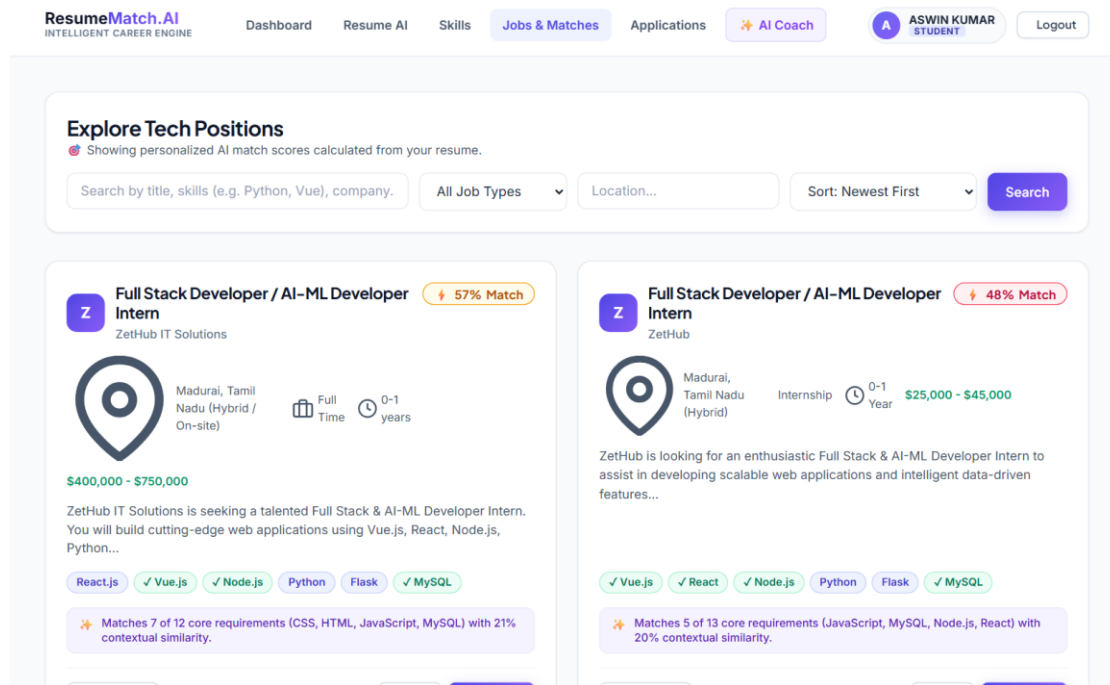


Fig 15: Student Job Board with Dynamic AI Match Percentages

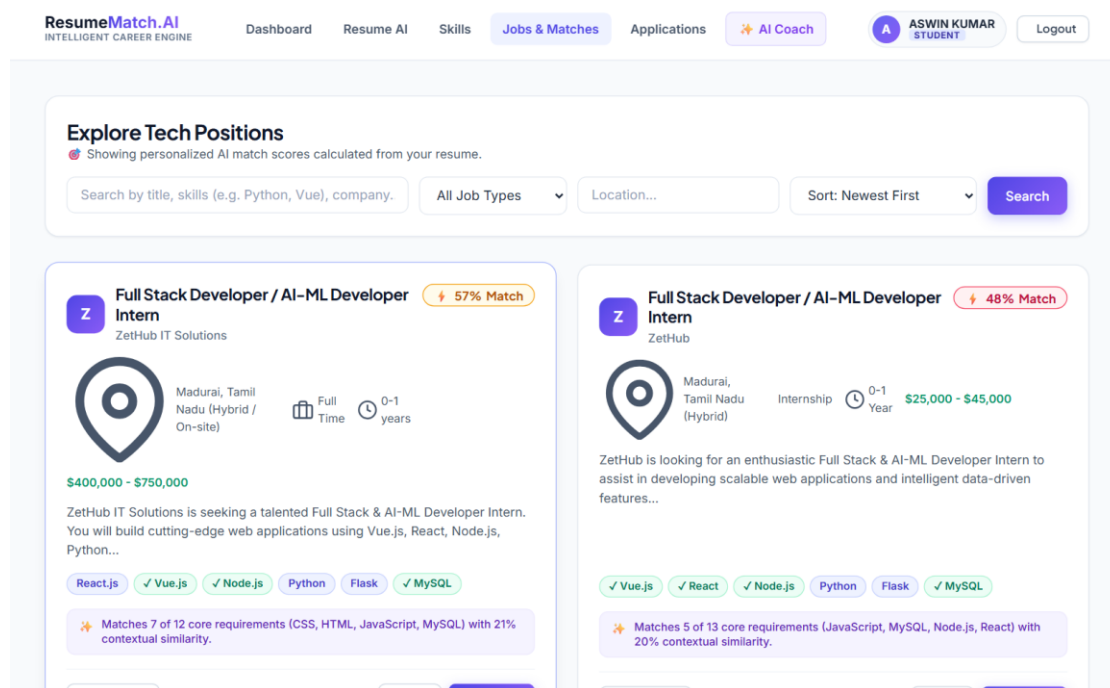


Fig 16: Explainable AI Job Compatibility Breakdown

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Dashboard

Resume AI

Skills

Jobs & Matches

Applications

AI Coach

ASWIN KUMAR

STUDENT

Logout

Explore Tech Positions

Showing personalized AI match scores calculated from your resume.

Search by title, skills (e.g. Python, Vue), company.

All Job Types

Location...

Sort: Newest First

Search

Z

Full Stack Developer / AI-ML Developer Intern

57% Match

ZetHub IT Solutions

Madurai, Tamil Nadu (Hybrid / On-site)

Full Time

0-1 years

\$400,000 - \$750,000

ZetHub IT Solutions is seeking a talented Full Stack & AI-ML Developer Intern. You will build cutting-edge web applications using Vue.js, React, Node.js, Python...

React.js

Vue.js

Node.js

Python

Flask

MySQL

Matches 7 of 12 core requirements (CSS, HTML, JavaScript, MySQL) with 21% contextual similarity.

Z

Full Stack Developer / AI-ML Developer Intern

48% Match

ZetHub

Madurai, Tamil Nadu (Hybrid)

Internship

0-1 Year

\$25,000 - \$45,000

ZetHub is looking for an enthusiastic Full Stack & AI-ML Developer Intern to assist in developing scalable web applications and intelligent data-driven features...

Vue.js

React

Node.js

Python

Flask

MySQL

Matches 5 of 13 core requirements (JavaScript, MySQL, Node.js, React) with 20% contextual similarity.

Fig 17: Successful Student Job Application Submission

ResumeMatch.AI

INTELLIGENT CAREER ENGINE

Dashboard

Resume AI

Skills

Jobs & Matches

Applications

AI Coach

ASWIN KUMAR

STUDENT

Logout

Application Tracker

Track the status of your submitted job applications in real-time.

Find More Jobs

No Applications Submitted

You have not applied for any roles yet. Browse recommendations and submit your first application!

Find Matching Jobs

Fig 18: Student Applied Jobs & Status Tracker

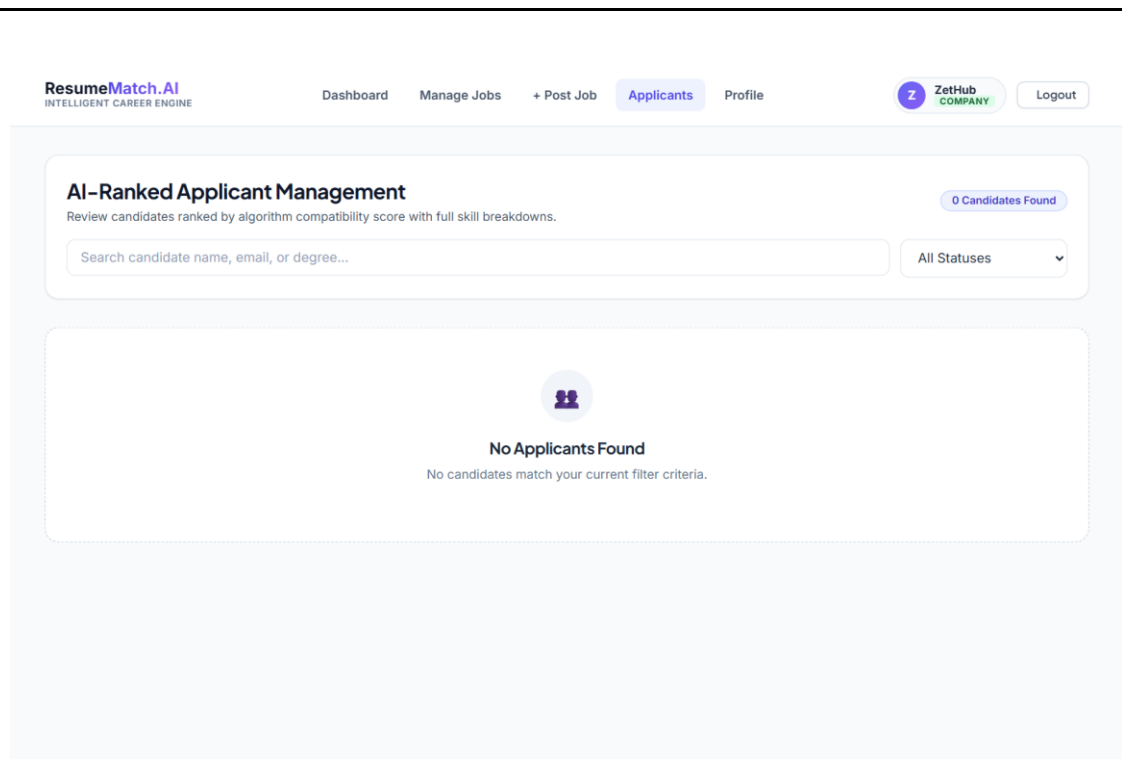


Fig 19: AI-Ranked Candidate List on ZetHub Portal

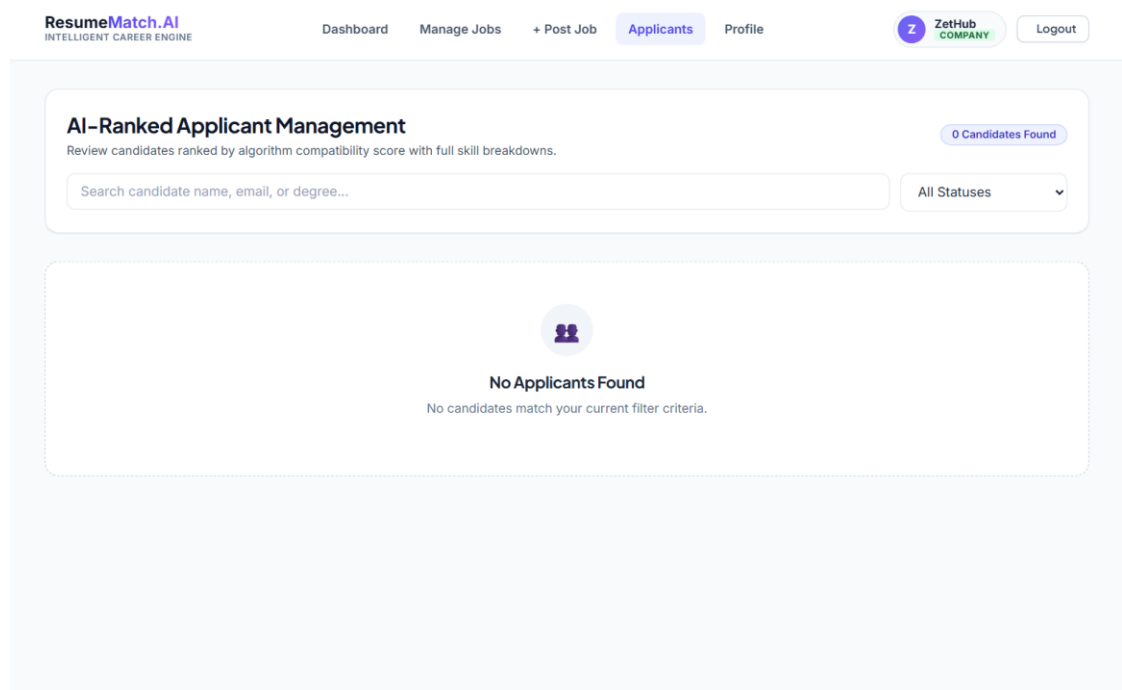


Fig 20: Candidate Shortlisting in Hiring Pipeline

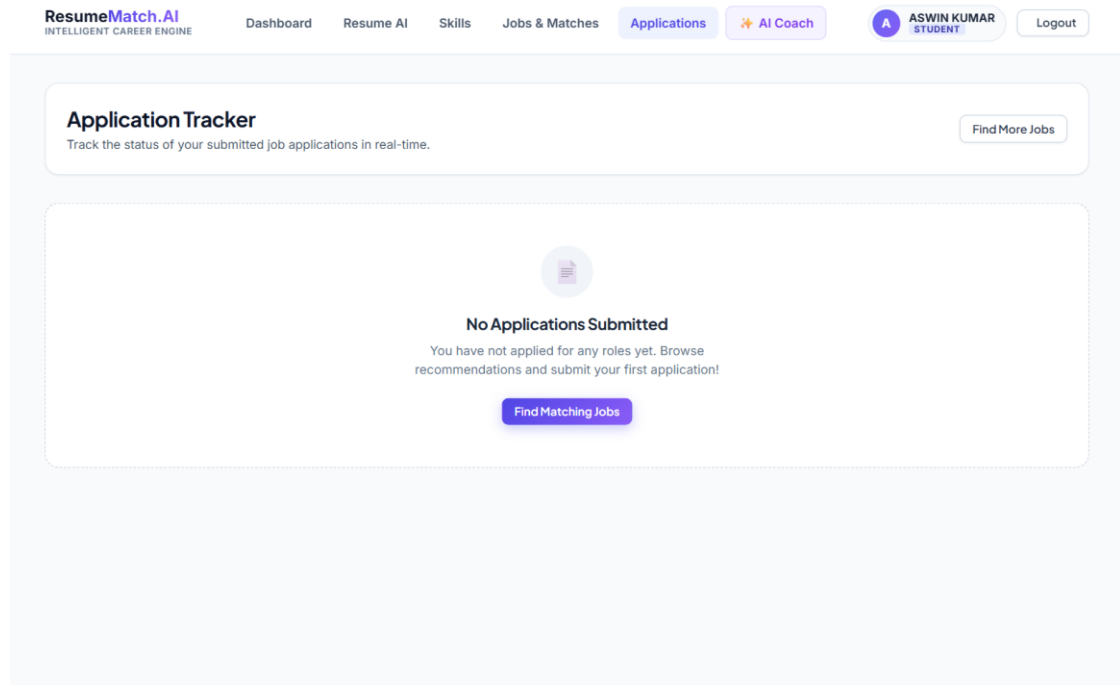


Fig 21: Updated Application Status Reflected to Student

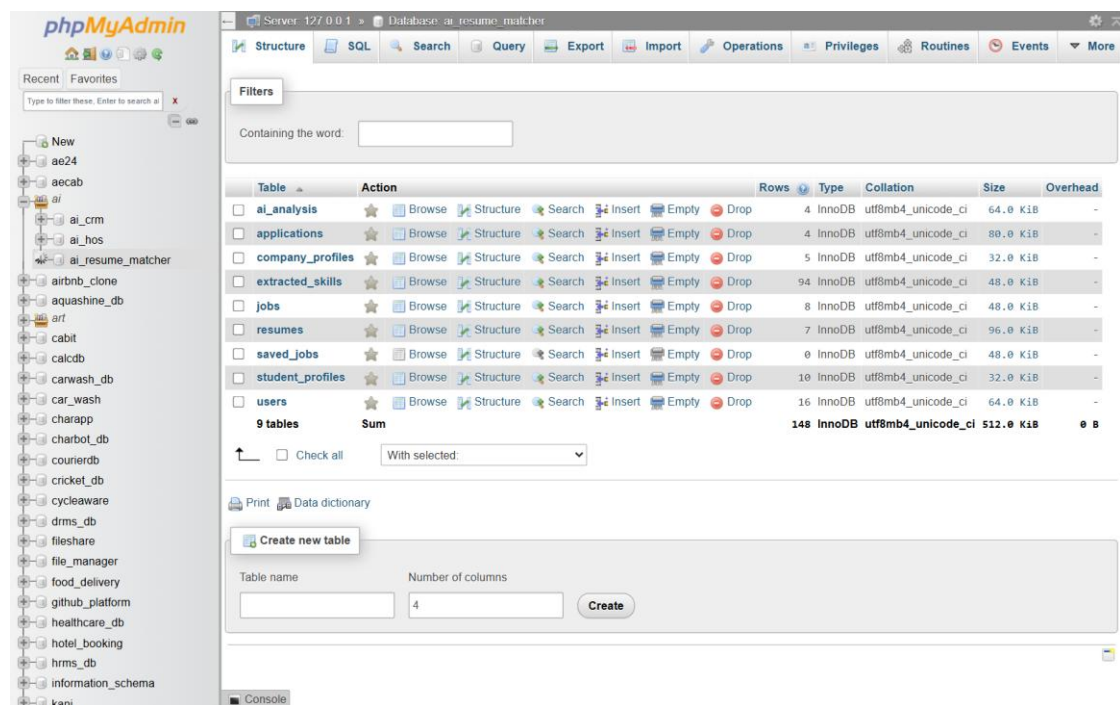


Fig 22: MySQL Database Structure in phpMyAdmin (ai_resume_matcher)

phpMyAdmin

Recent Favorites

Type to filter these. Enter to search at

New

- ae24
- ae24
- ae24
- ai
- ai_crm
- ai_hos
- ai_resume_matcher
- New
- ai_analysis
- applications
- company_profiles
- extracted_skills
- jobs
- resumes
- saved_jobs
- student_profiles
- users

airbnb_clone

aquashine_db

art

cabit

calcdb

carwash_db

car_wash

charapp

charbot_db

counerd

cricket_db

cvcleaware

Server: 127.0.0.1 » Database: ai_resume_matcher » Table: users

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Showing rows 0 - 15 (16 total, Query took 0.0016 seconds)

SELECT * FROM `users`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		id	name	email	password_hash	role	status
☐	Edit Copy Delete	1	System Administrator	admin@airesume.com	script:32768:8:1\$OfsLEn3lg4pS1JveSeb4615e163f6eae...	admin	active
☐	Edit Copy Delete	2	TechNova Solutions	hr@technova.com	script:32768:8:1\$re2uFNvGmBlgsuk\$3c53fa5cb6b063a2...	company	active
☐	Edit Copy Delete	3	ZetHub	careers@zetHub.com	script:32768:8:1\$0yFMnZmDQ5qAxDKN\$C03fda5ad732d75b...	company	active
☐	Edit Copy Delete	4	CloudCore Systems	jobs@cloudcore.com	script:32768:8:1\$VQZJTxFT2lntIDh\$D9877076b0679c26...	company	active
☐	Edit Copy Delete	5	DataSphere AI	talent@datasphere.ai	script:32768:8:1\$X5ur9vmJFUThm\$888003a214632ba8...	company	active
☐	Edit Copy Delete	6	Alex Rivera	alex.student@example.com	script:32768:8:1\$CQyfwH5z4IqGu6r7\$5d3299e3252bf1c2...	student	active
☐	Edit Copy Delete	7	Integration Test Student	test_student_04d1e4@example.com	script:32768:8:1\$3siZ11AEYxzD3yvo\$4de45ac4be6ac0ac...	student	active
☐	Edit Copy Delete	8	Integration Test Student	test_student_2cb149@example.com	script:32768:8:1\$TFn7TnFTgptVikp\$99792988ba4b57dd...	student	active
☐	Edit Copy Delete	9	E2E Candidate	e2e_student_d8a5be45@test.com	script:32768:8:1\$Q5bDe1RTKbYyxUM\$ea3ce9bd04696fe8...	student	active
☐	Edit Copy Delete	10	Integration Test Student	test_student_b313ad@example.com	script:32768:8:1\$e3tYJYcLWb0C7uaZ8\$2dc55c007eefef32...	student	active
☐	Edit Copy Delete	11	E2E Candidate	e2e_student_c86bc46c@test.com	script:32768:8:1\$ZqXTO5vLdJmUDUIG\$5b23c18cd91bd17f...	student	active

Console

Fig 23: MySQL Users Table Data Records

phpMyAdmin

Recent Favorites

Type to filter these. Enter to search at

New

- ae24
- ae24
- ae24
- ai
- ai_crm
- ai_hos
- ai_resume_matcher
- New
- ai_analysis
- applications
- company_profiles
- extracted_skills
- jobs
- resumes
- saved_jobs
- student_profiles
- users

airbnb_clone

aquashine_db

art

cabit

calcdb

carwash_db

car_wash

charapp

charbot_db

counerd

cricket_db

cvcleaware

Server: 127.0.0.1 » Database: ai_resume_matcher » Table: jobs

Browse Structure SQL Search Insert Export Import Privileges Operations More

Showing rows 0 - 7 (8 total, Query took 0.0024 seconds)

SELECT * FROM `jobs`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		id	company_id	title	description	requirements	skills	preferred_skills	location	job_type	experie
☐	Edit Copy Delete	1	2	Full Stack Python & Vue Developer	We are looking for a versatile Full Stack Developer...	Bachelor in CS or related experience. Strong profi...	Python, Flask, Vue.js, JavaScript, MySQL, REST API...	Docker, Tailwind CSS, Redis, TypeScript	San Francisco, CA (Hybrid)	Full Time	1-3 yea
☐	Edit Copy Delete	2	2	Frontend Engineer (Vue 3 / TypeScript)	Join our frontend engineering team to craft blazin...	Strong command of Vue.js, modern JavaScript/TypeSc...	Vue.js, JavaScript, TypeScript, HTML, CSS, REST AP...	Tailwind CSS, Vite, Jest, Figma	San Francisco, CA	Full Time	2-4 yea
☐	Edit Copy Delete	3	3	AI / Machine Learning Intern	Exciting opportunity for a motivated AI/ML intern ...	Currently enrolled or recent graduate in CS/Data S...	Python, Machine Learning, NLP, scikit-learn, NumPy...	Deep Learning, PyTorch, TensorFlow, Flask (Remote)	New York, NY (Remote)	Internship	0-1 yea
☐	Edit Copy Delete	4	3	Backend Python Engineer	Architect and scale backend services, microservices, with Python, ...	2+ years developing backend services	Python, Flask, MySQL, REST API, JWT, SQL, Git, Lin...	FastAPI, Docker, Redis, Celery	New York, NY	Full Time	2-5 yea

Console

Fig 24: MySQL Published Jobs Table Data

phpMyAdmin

Server: 127.0.0.1 » Database: ai_resume_matcher » Table: applications

Showing rows 0 - 3 (4 total. Query took 0.0007 seconds.)

SELECT * FROM `applications`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		id	job_id	student_id	resume_id	match_score	status	cover_note	applied_at	updated_at
☐	Edit Copy Delete	1	1	9	1	70.8	Applied	Excited to apply for this role!	2026-09-26 19:28:08	2026-09-26 19:28:08
☐	Edit Copy Delete	2	1	11	2	70.8	Shortlisted	Excited to apply for this role!	2026-09-26 19:28:32	2026-09-26 19:28:33
☐	Edit Copy Delete	3	1	13	3	70.8	Shortlisted	Excited to apply for this role!	2026-09-26 20:01:15	2026-09-26 20:01:15
☐	Edit Copy Delete	4	1	14	5	44	Applied	...Id	2026-09-26 20:47:24	2026-09-26 20:47:24

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Console this SQL query

Fig 25: MySQL Job Applications Table Records

RESULT:

The system successfully:

- Provides secure multi-role registration and authentication for Students, Companies, and Administrators.
- Parses unstructured resumes in PDF, DOCX, and TXT formats and extracts candidate profiles.
- Identifies and categorizes technical skills dynamically using an NLP taxonomy engine.
- Calculates an ATS health score with actionable suggestions for resume optimization.
- Enables employers (ZetHub) to create, publish, and manage structured job vacancies.
- Executes a TF-IDF and Cosine Similarity matching pipeline to compute exact candidate-job compatibility.
- Allows seamless student application submissions and company hiring pipeline management (Shortlisting).
- Ensures ACID-compliant persistent relational storage using MySQL and XAMPP.

Thus, the mini project effectively integrates Artificial Intelligence, Machine Learning, Natural Language Processing, and MySQL database technologies to automate recruitment, simplify candidate evaluation, and optimize student-job matching.

REFERENCES:

1. Scikit-learn Machine Learning Library Documentation – <https://scikit-learn.org/stable/>
2. Python Official Documentation – <https://docs.python.org/3/>
3. Flask Web Development Framework Documentation – <https://flask.palletsprojects.com/>
4. Vue.js Progressive JavaScript Framework – <https://vuejs.org/guide/>
5. MySQL 8.0 Reference Manual – <https://dev.mysql.com/doc/>
6. PyPDF & python-docx Documentation – <https://pypdf.readthedocs.io/>
7. W3Schools & MDN Web Docs (JavaScript, CSS, HTML5) – <https://developer.mozilla.org/>